TIME SERIES FORECASTING INTRODUCTION

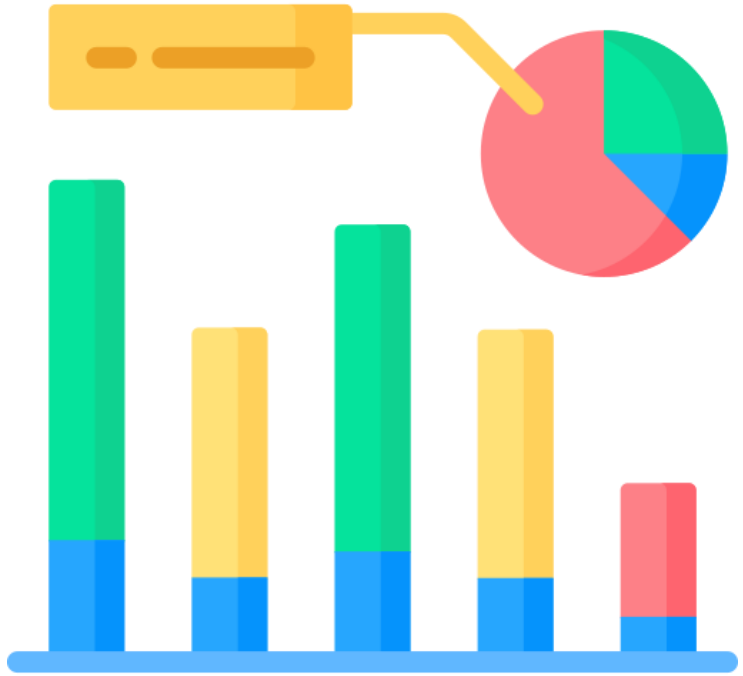
LECTURER. THONG LAKSMEY

DEPARTMENT OF APPLIED MATHEMATICS AND STATISTICS

TIME SERIES FORECASTING INTRODUCTION

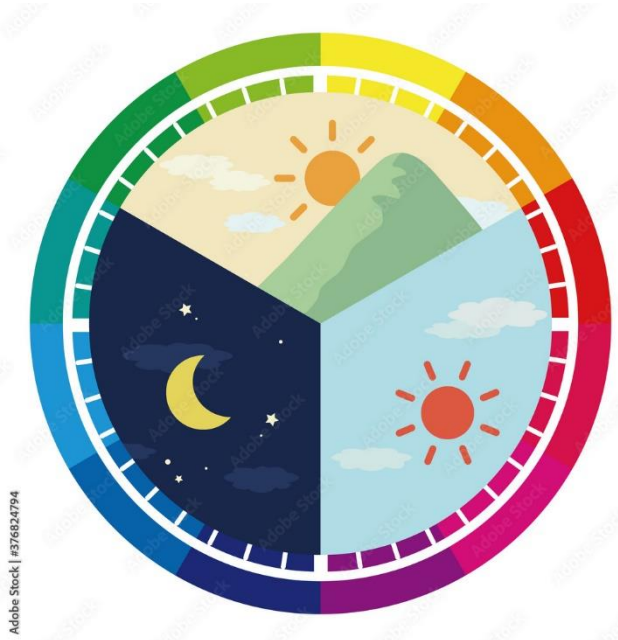
- Time Series Forecasting is the use of a statistical model to predict future values, based one past results. Any value that can be tracked and collected over time.
- Think of annual population data, a company's daily stock price or Quarterly Sale Figures.





TIME SERIES MODEL

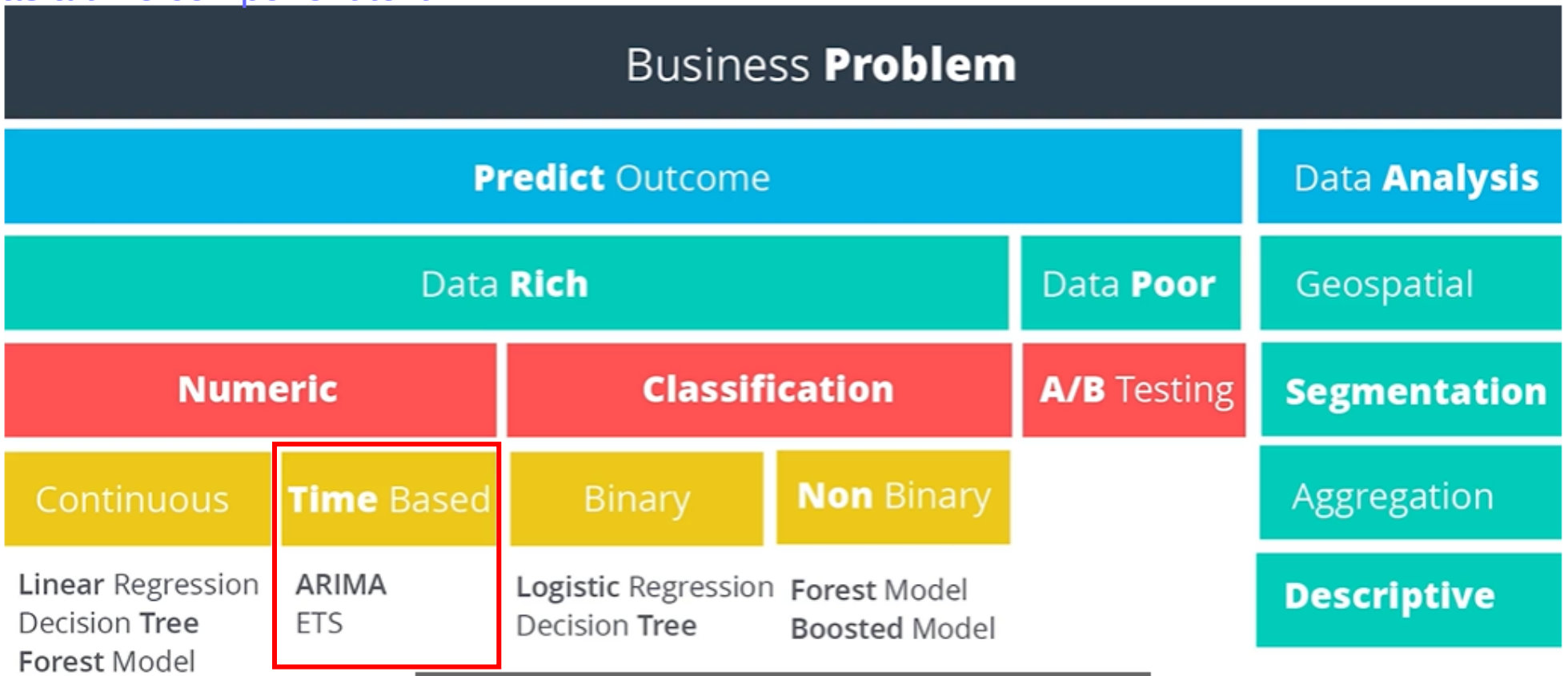
- How do I define time ? What are trend ? What are seasonal Pattern ?
- Time Series Forecasting Model
- Exponential Smoothing (ETS) and Autoregressive Integrated Moving Average or ARIMA Models.
- Briefly look at where time series forecasting fits, In the Problem Solving with Advanced Analytics course, you're introduced to Methodology Map which is used to determine which methodology is best for a particular analysis
- So, with our business problem, we need to identify if we're doing data analysis or trying to predict an outcome.





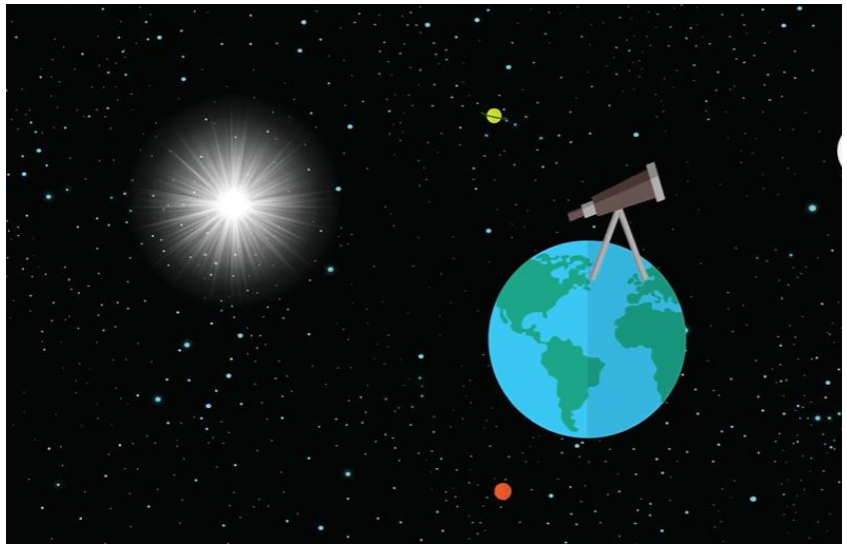
TIME SERIES FORECASTING INTRODUCTION

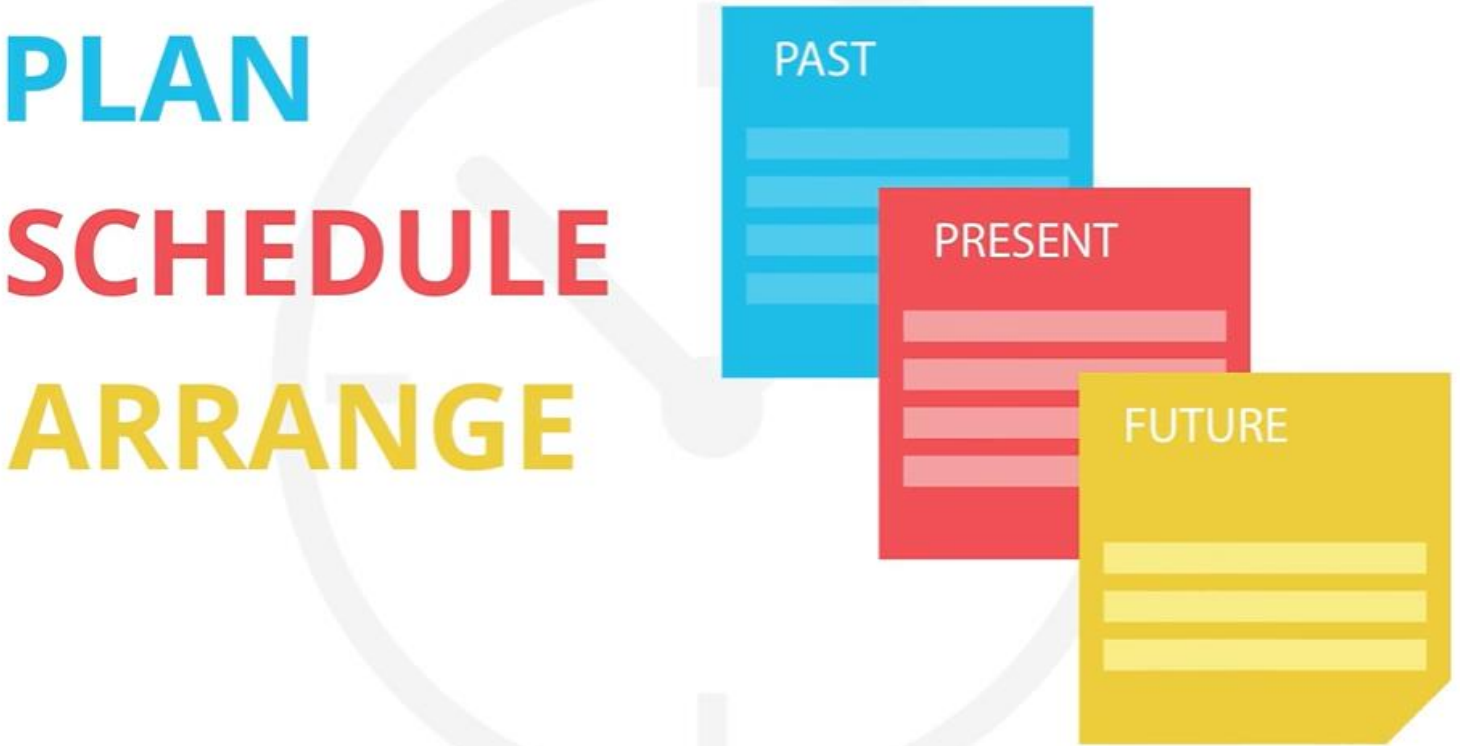
Example : Year over Year population a stock's daily closing price and lastly, our quarterly sales figures. All these are numeric variable. The data for these examples are collected by year, day, and quarter, respectively. So, each observation has a time component to it.



TIME SERIES FORECASTING INTRODUCTION

- Ever since man kind first noticed the regular movement of the sun and the starts, we have wondered about the passage of time. Our ancestors first recorded the phases of the moon some 30,000 years ago, and recording time has been a way by which humanity has represented the progress of civilization.
- It's how we plan, schedule, or arrange when something should happen or be done. It's the events in our past, present, and future.





TIME SERIES FORECASTING INTRODUCTION

- **The first defining characteristic** of a time series is a list of observations where the ordering matters.
- **Ordering is very important because there is a dependency on time and changing the order could change the meaning of the data.**
- Now, to accurately forecast future values, we'll need each measurement of data to be taken across sequential and equal intervals, and with each time unit having at most one data point.

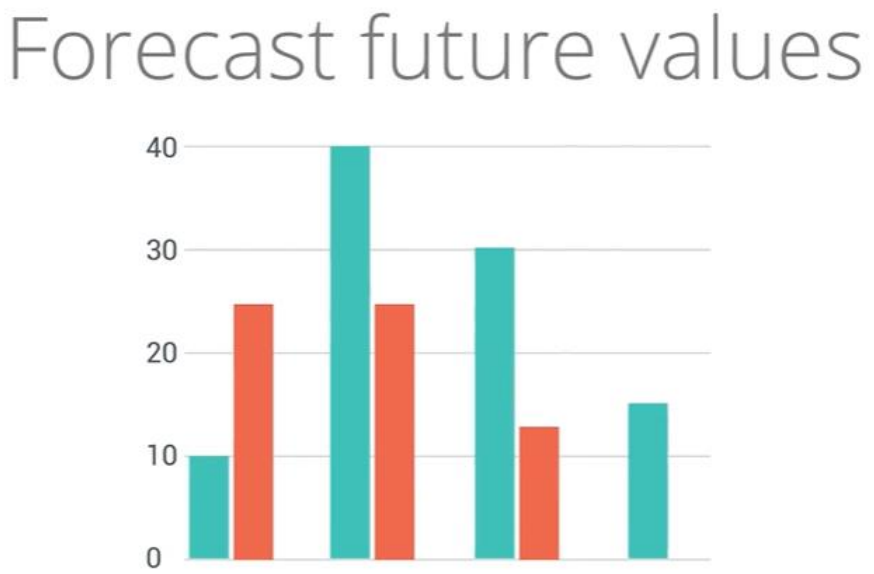




TIME SERIES FORECASTING INTRODUCTION

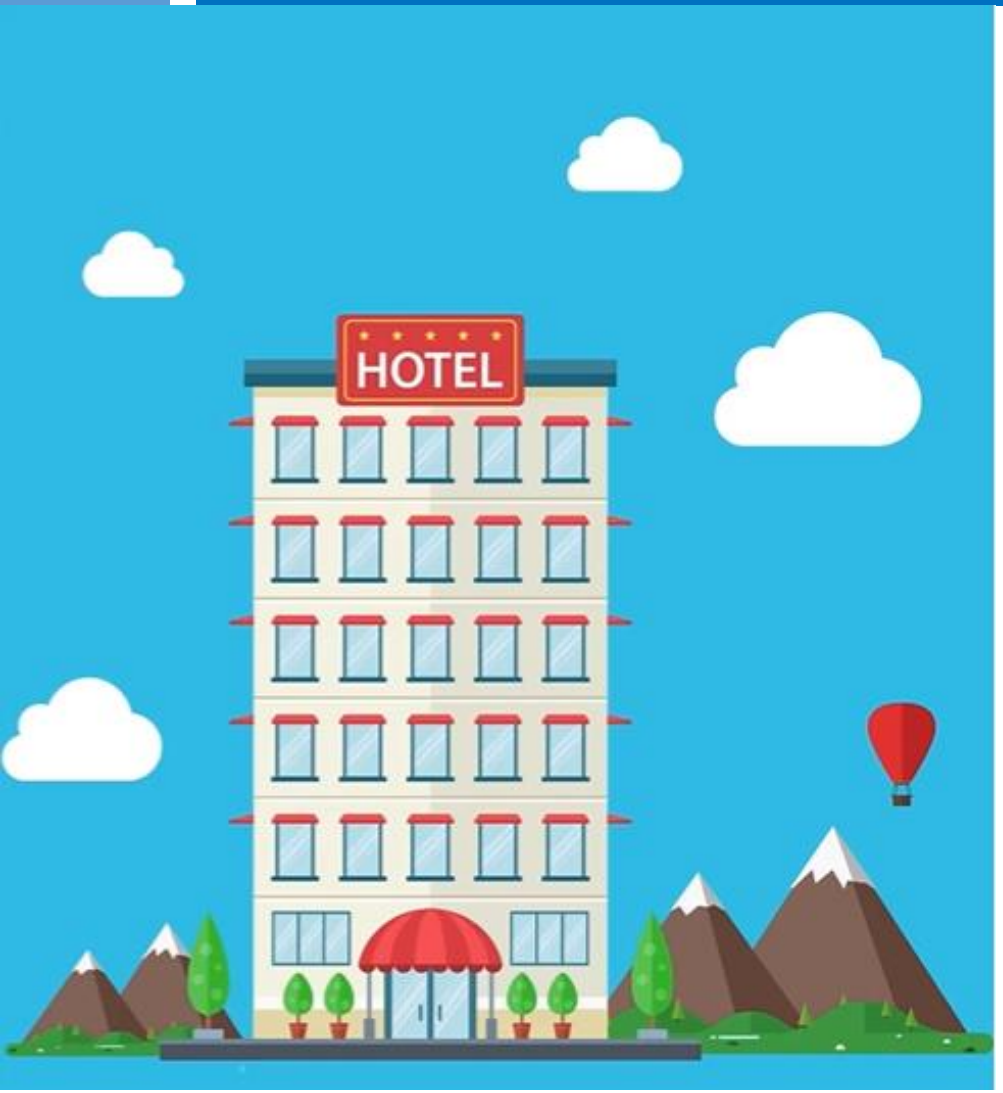
- Once we have collected our data, as a business analyst, we have two objectives in mind. **First, identifying the patterns represented by the sequence of observations**, and **second, forecasting or predicting future values of the time series.**
- The patterns we observe will tell us the story of how time of our business interacts with time.







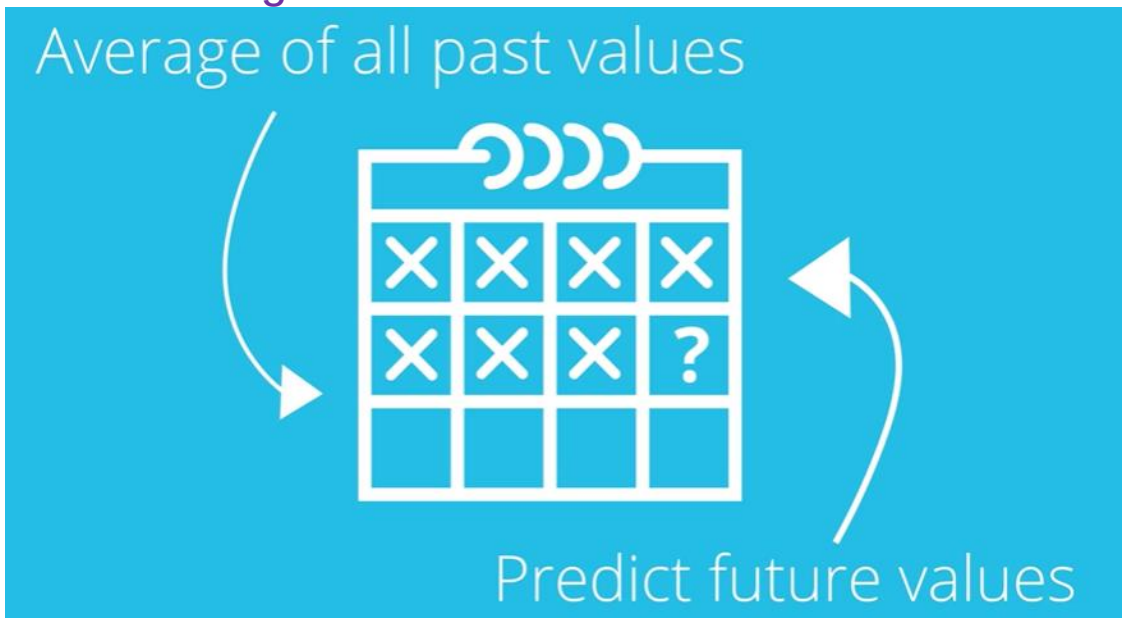
TIME SERIES FORECASTING INTRODUCTION



- To better understand and apply these methods, we'll examine a business problem surrounding how management **at Bates Hotel,** a mountain resort walking distance from the best slopes in town, are able to prepare for their year by **forecasting the number of room bookings they expect each month.**
- This information can help management make informed decisions on staffing, hospitality arrangements, and pricing for rooms.
- They have tasked their business analysts, you, to forecast bookings so they can make the necessary preparations.
- Historical bookings data contains **monthly data from the past ten years**, and the forecast should contain monthly bookings for the next six months.
- Your job will to be use to the historical bookings data from years past. to one, investigate and clean the data.
- Two, determine trend and seasonal components, three, apply findings to an **ARIMA and ETS model in Alteryx**,
- lastly, forecast bookings for the next 6 months.

SIMPLE FORECASTING METHOD

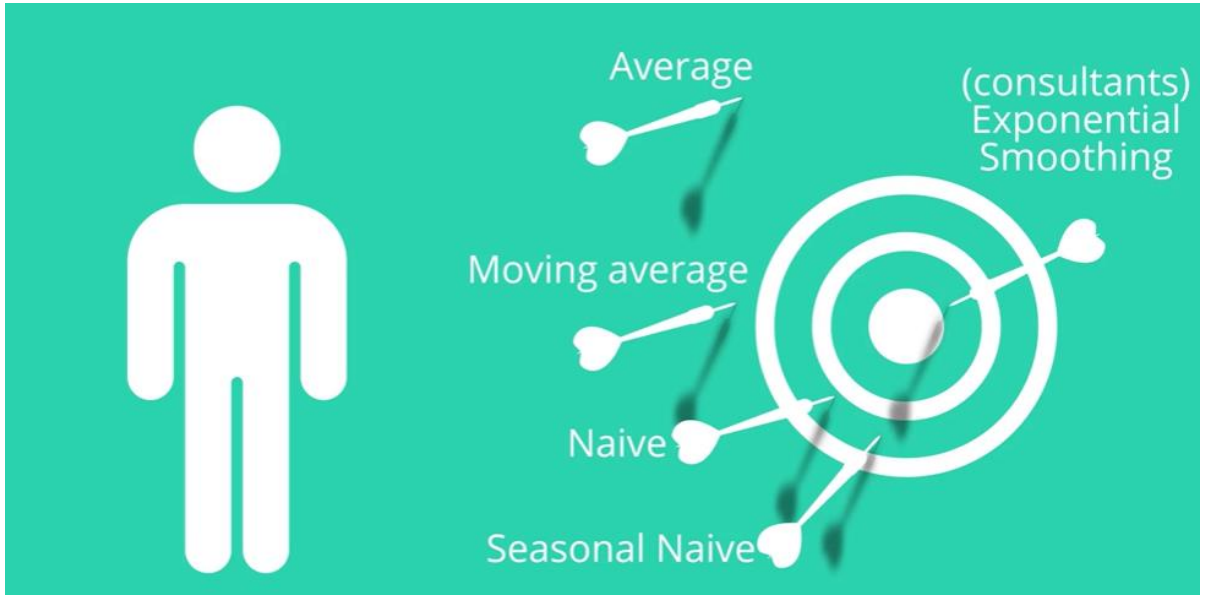
- Some forecasting methods are very simple and occasionally effective. But they do have their faults. Let's take a look at Sam, the simple forecaster, who previously worked in your position at Bates Hotel and how he used simple forecasting methods to forecast bookings.
- When Sam was first tasked with forecasting bookings for Bates Hotel, he didn't know how to use a sophisticated predictive model to add early forecast results. Nor did Sam want to take the time to learn.
- **Instead, Sam used simple forecasting methods so he could hit the slopes before they closed at the end of the day.**
- Sam's first attempt at forecasting was to use an **average method**. Here, the forecast of all future values are equal to the average of historical data.



SIMPLE FORECASTING METHOD

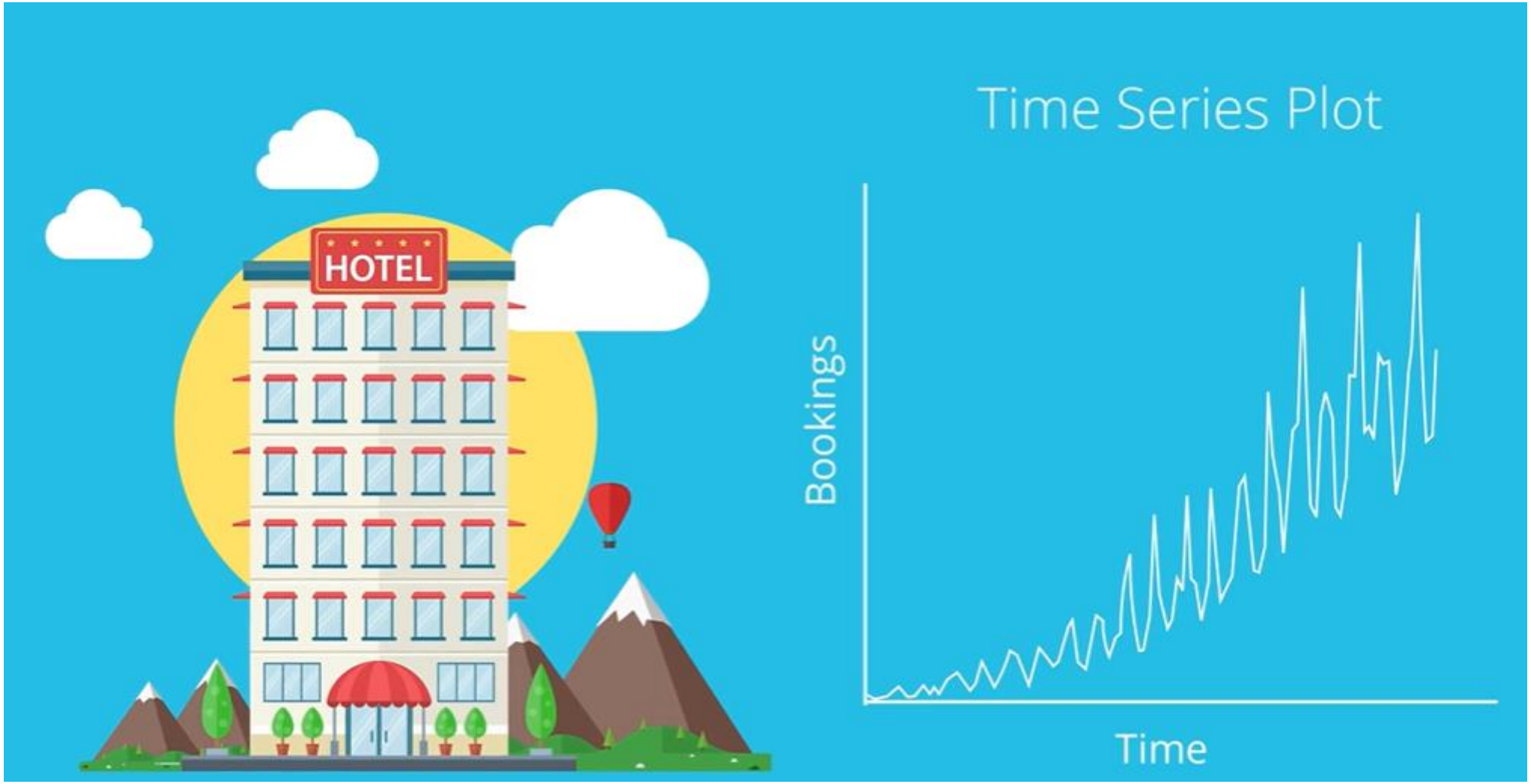
- **Average Method**
- **The best predictor of what will happen tomorrow is the average of everything that has happened up until now.**
- For Sam and Bates Hotel management, this was not the case. Using our Sam the simple forecaster CSV, we can calculate what Sam forecasted using this method.
- As you can see, Sam missed his mark by over 9,000 bookings. This method uses the average o fa time series over a defined number of time periods.
- Sam figured that the last 12 months of bookings would be a good moving average to base his next forecast. Again, Sam was wrong, missing by almost 3,000 bookings.
- I decided to give another method a try since he's been unsuccessful with his previous two forecasts.
- This time he tried the naive method. In this method all forecasts are simply set to be the value of the last observation. Sam is getting closer yet still off by over 1,000 bookings.





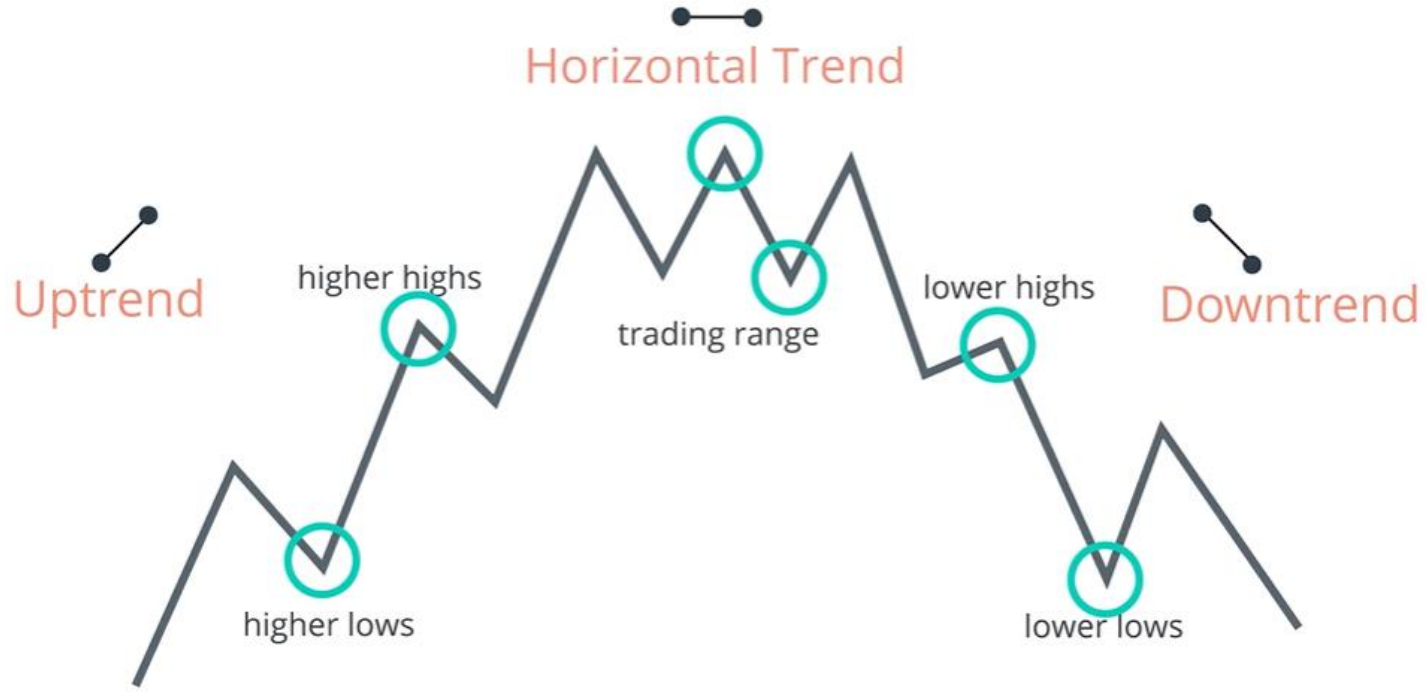
TIME SERIES COMPONENT

- A time series plot shows a graphical presentation of the relationship between **time and the time series target variable.**
- Time is on the horizontal axis and the target variable's value sare shown on the vertical axis.



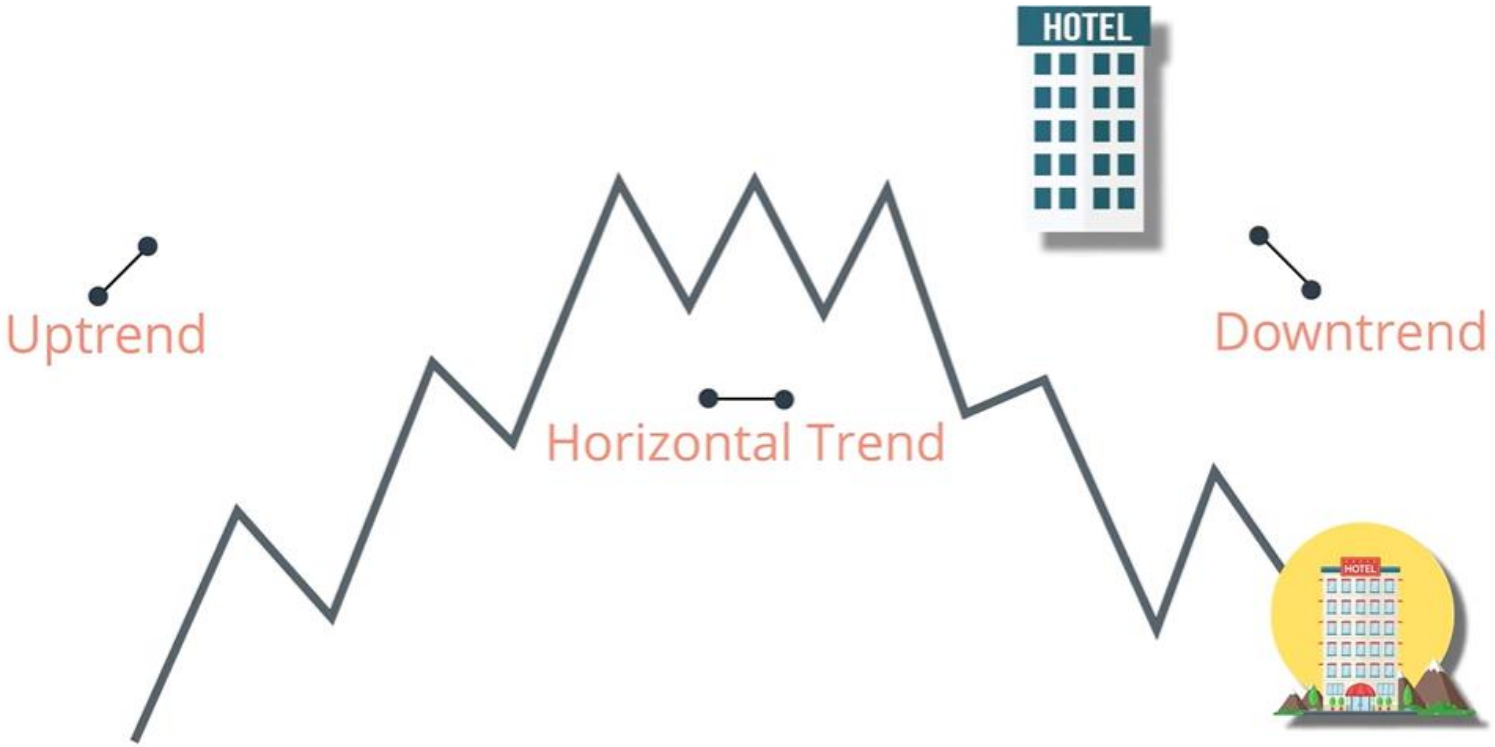
TREND

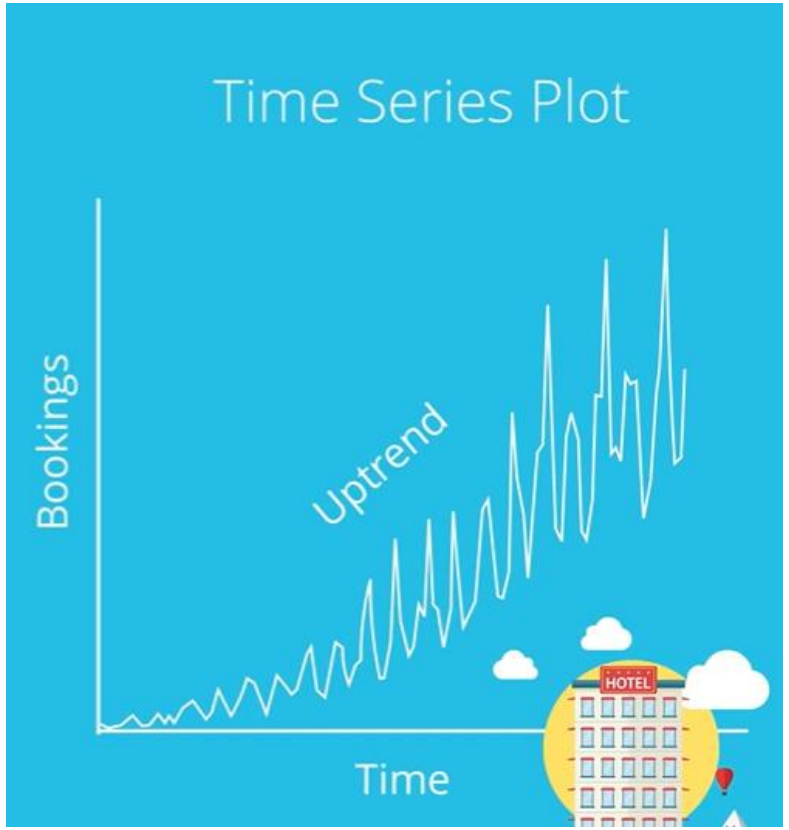
- In time series analysis , a trend is a gradual shift or movement to relatively higher or lower values over a long period of time. When our trend pattern exhibits a general direction that is upward, where there are higher highs and higher lows, we call this an uptrend.
- When our trend pattern exhibits a general direction that is down, where there are lower highs and lower lows, we call this a downtrend.
- where there are lower highs and lower lows, we call this a downtrend. an Uptrend to a Downtrend. If there were no trend, we call it a Horizontal or stationary trend. Each trend we see in a timeseries is call the trend cycle.



TREND

- **When looking at the plot of the Bookings data from Bates Hotel, we see that Bates Hotel continues to increase their bookings year over year, creating an uptrend in the time series**.
- If Bates Hotel reaches capacity, then bookings will trend sideways. And should a competing hotel open down the street and steal their bookings, Bates would experience a downtrend.



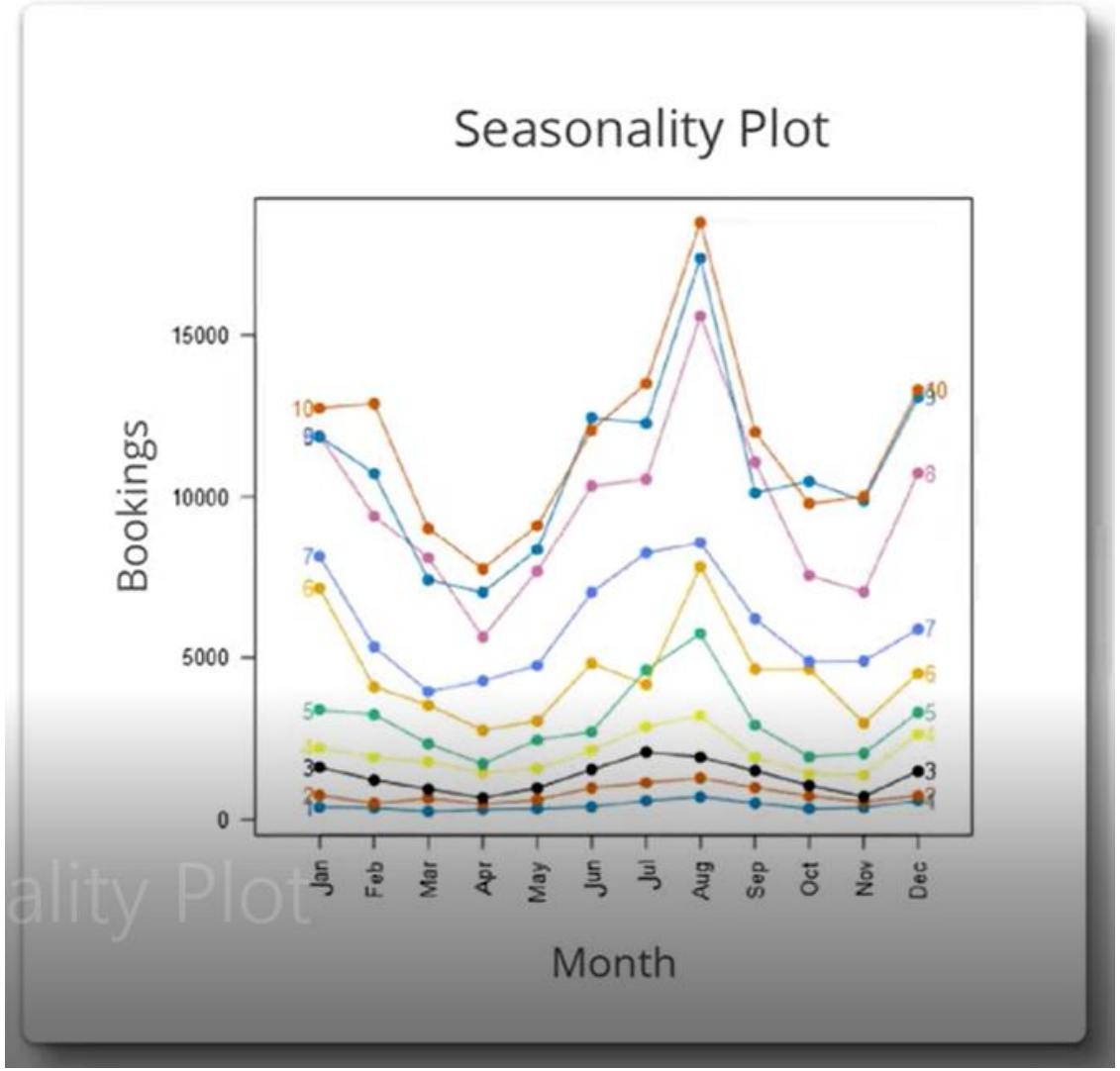


SEASONALITY

- A time series that exhibits a repeating pattern at fixed intervals of time within a one year period is said to have a seasonal pattern or seasonality.
- Seasonality is a common pattern seen across many different kinds of time series. For example, if you live in a climate with cold Winters and warm Summers your home's air conditioning costs probably rise in the summer and fall in the winter. You would reasonably expect the seasonality of your air conditioning costs to recur every year.
- Likewise, a company that sells heavy coats would see sales jump in the Winter but drop in the Summer. Companies that understand the seasonal patterns of their business can time inventories, staffing, and other decisions that coincide with the expected change in business.
- The US postal service hiring temporary drivers and workers during the holiday season shows how a company adapts to meet a change caused by seasonal demand.

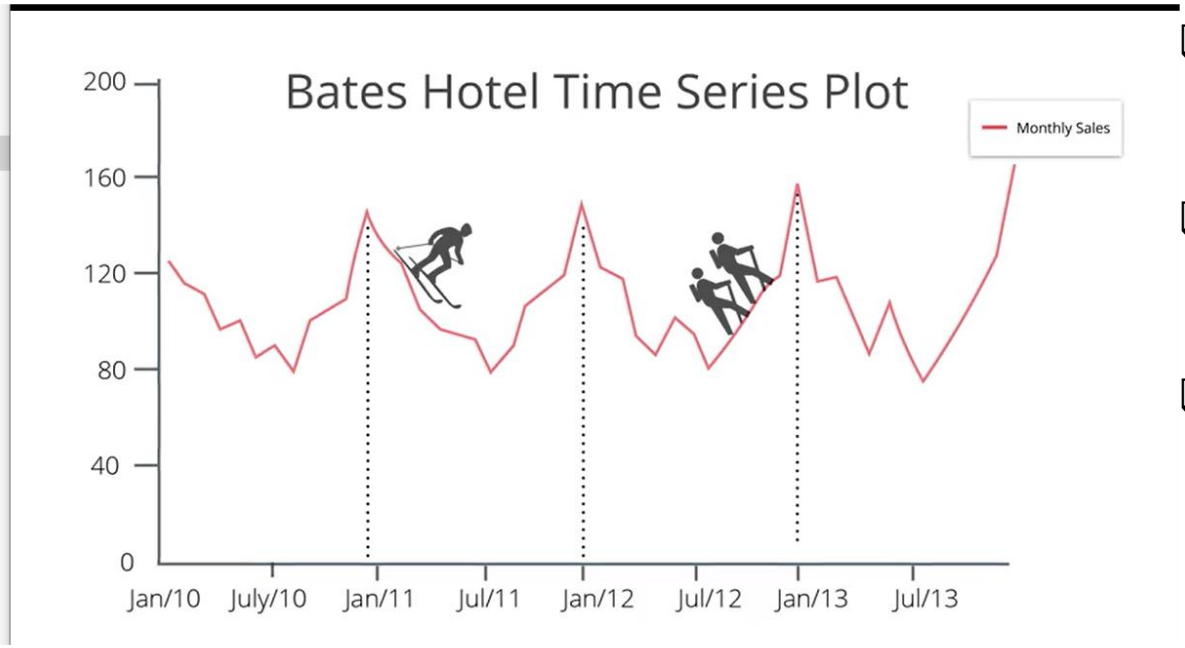


SEASONALITY PLOT



- All time series plots, the vertical axis is our target variable,
- The horizontal axis shows our seasonality. For example, with monthly data, all the monthly values are plotted in chronological order with each month on the horizontal axis.
- **The numbers one through ten represent the ten years of data that we have seen in the time series.**

SEASONALITY PLOT



- When looking at this plot with the Bates Hotel time series, we can tell our results during winter and summer months are significantly higher.
- These findings are not surprising, **considering most people go to the hotel during these months to take advantage of the skiing and hiking.**
- **Since this pattern repeats each year at the same time intervals,** we can safely say that our timeseries contains seasonality. Another noticeable characteristic of the Bates Hotel booking seasonality plot, is that the magnitude of bookings increases year after year.

CYCLICAL PATTERN

- What if we see a pattern in our data, that doesn't occur within the same calendar year? Is that still a seasonal pattern?
- A cyclical pattern exists, **when data exhibits rises and falls that are not of a fixed period**. Think of business cycles which usually last several years, but where the length of the current cycle is unknown beforehand.
- In finance, times of expansion and recession in the stock market reveal cyclical patterns. **A cyclical uptrend is referred to a bull market, while a cyclical downtrend is referred to as a bear market. These patterns in the general market occupy multiple years, and don't have a repeating pattern within each year.**
- Many people confuse cyclical behavior with seasonal behavior, but they are quite different.
- **If the fluctuations are not of a fixed period, then they are cyclical.**
- If the period is unchanging and associated with some aspect of the calendar, then the pattern is seasonal.
- In general, with cyclical patterns, the average length of cycles is longer than the length of the seasonal pattern.
- And the magnitude of cycles tends to change more than the magnitude of seasonal patterns.
- Also, cyclical patterns are much harder to predict as well. The decline in stock markets is often sudden and violent, all coming as a surprise to most.

TIME SERIES FORECASTING QUIZ

1. A time series is a sequence of data points with the follow characteristics (choose all that apply):
   a. Random intervals of time between each data point
   b. Covers a continuous time interval
   c. Equal spacing between every two consecutive measurements
   d. Is considered data poor
   e. Each time Unit within the time interval has at most one data point

2. We would use time series forecasting to predict values for which following business situations (select all that apply):
   a. Monthly beach bike rentals
   b. A stock's daily closing value
   c. Amount of time to close a sale cycle
   d. Demographic analysis on trade areas surrounding a toy store

TREND QUIZ

1. **Match the above time series to its corresponding trend.**

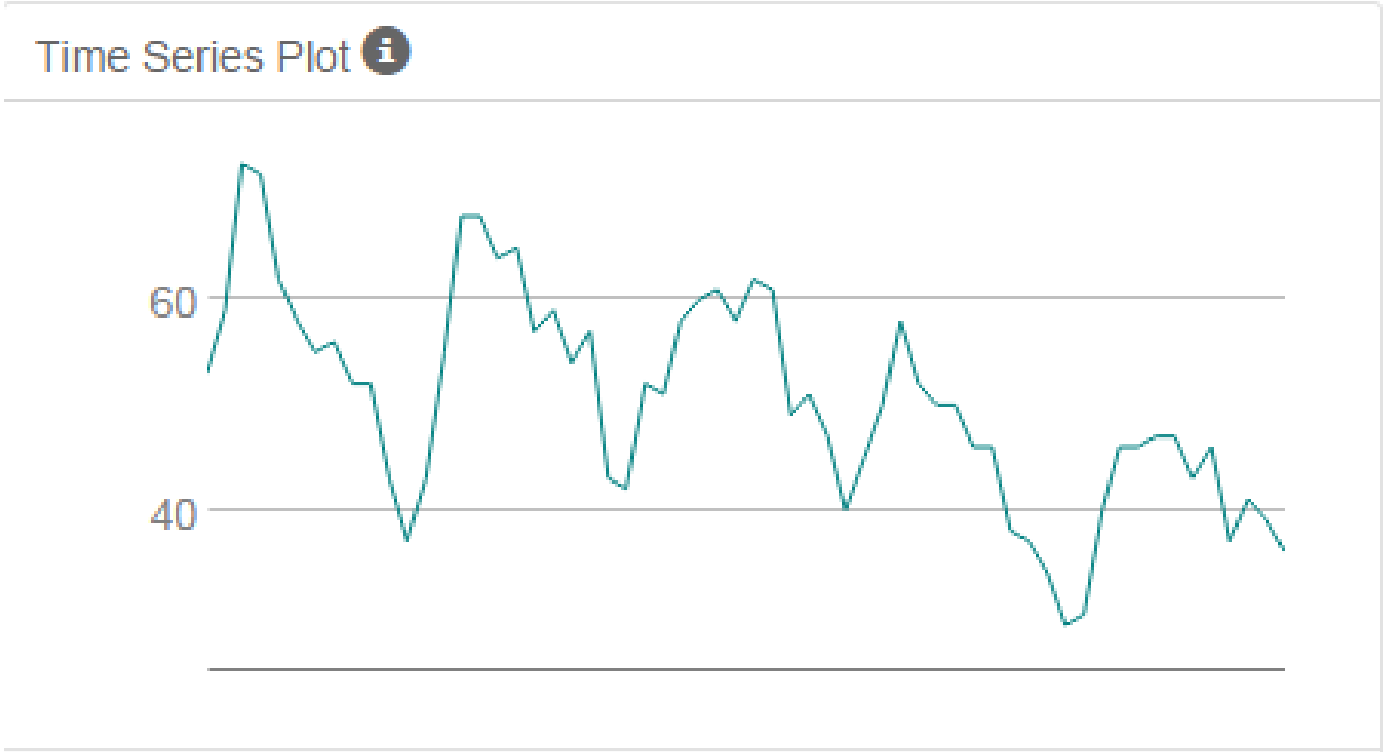
    a. Uptrend

    b. Downtrend

    c. Horizontal (no trend)

    d. Downtrend to Uptrend

    e. Uptrend to Downtrend



TREND QUIZ

2. Match the above time series to its corresponding trend.

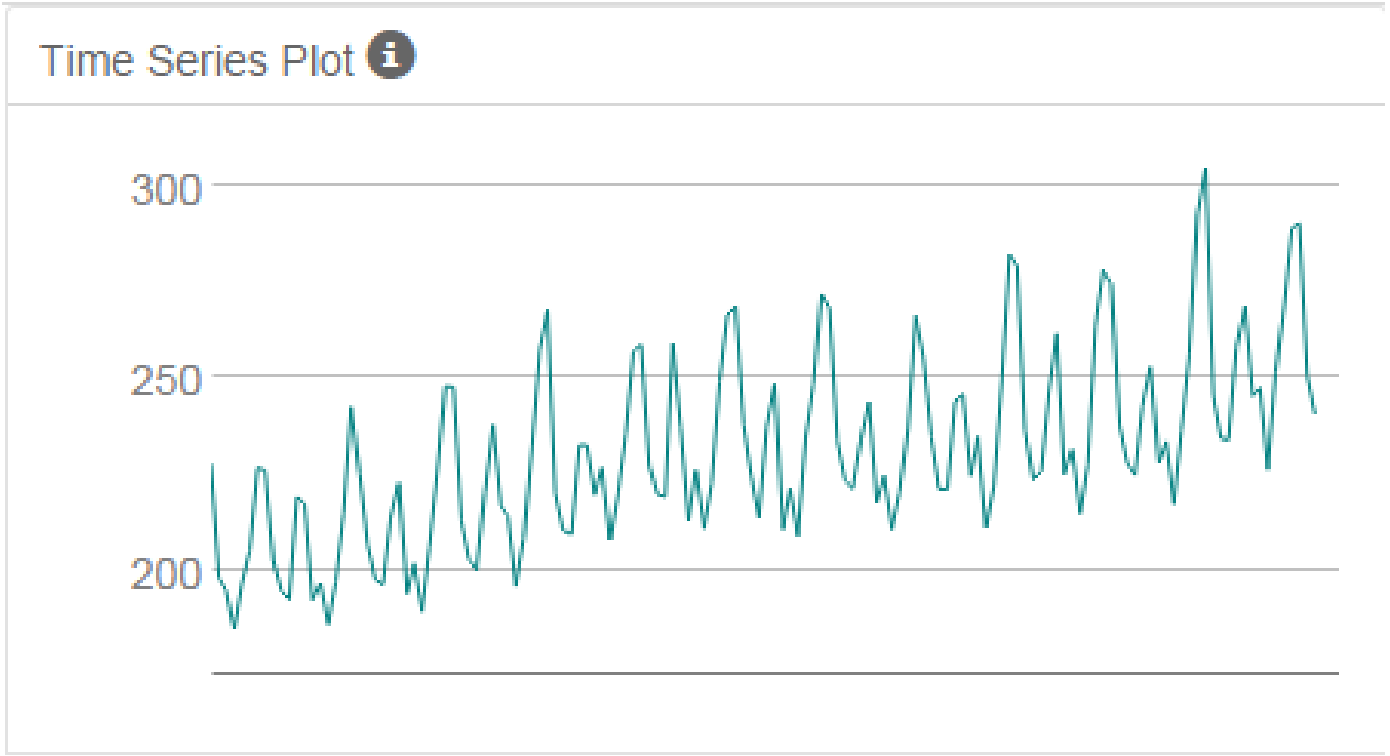
a. Uptrend

b. Downtrend

c. Horizontal (no trend)

d. Downtrend to Uptrend

e. Uptrend to Downtrend



TREND QUIZ

3. Match the above time series to its corresponding trend.

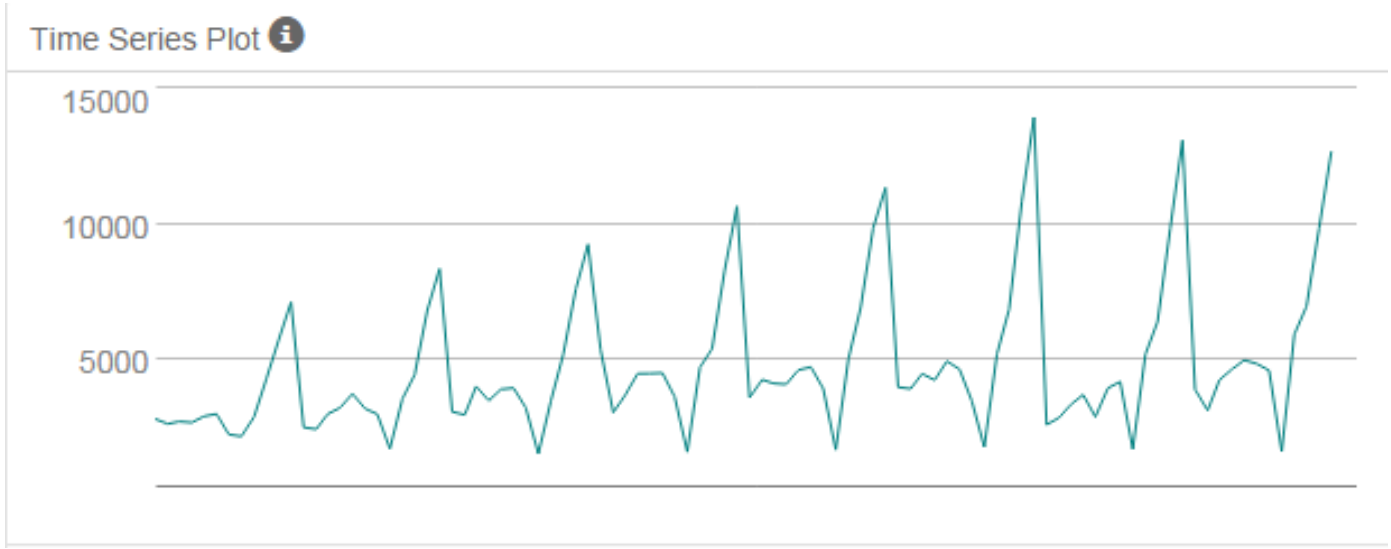
a. Uptrend + seasonality

b. Downtrend

c. Horizontal (no trend)

d. Downtrend to Uptrend

e. Uptrend to Downtrend



TREND QUIZ

4. Match the above time series to its corresponding trend.

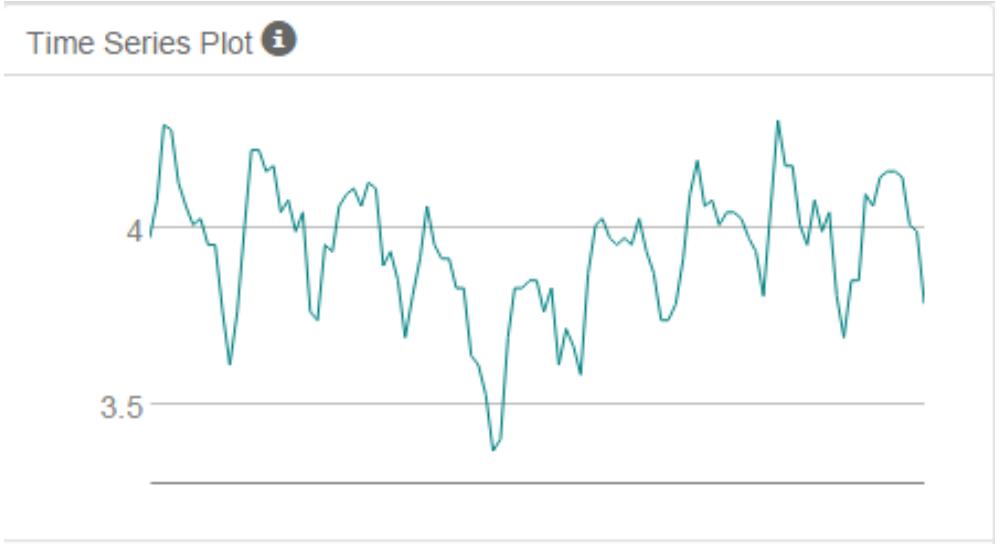
a. Uptrend

b. Downtrend

c. Horizontal Trend

d. Downtrend to Uptrend

e. Uptrend to Downtrend



TREND QUIZ

5. Match the above time series to its corresponding trend.

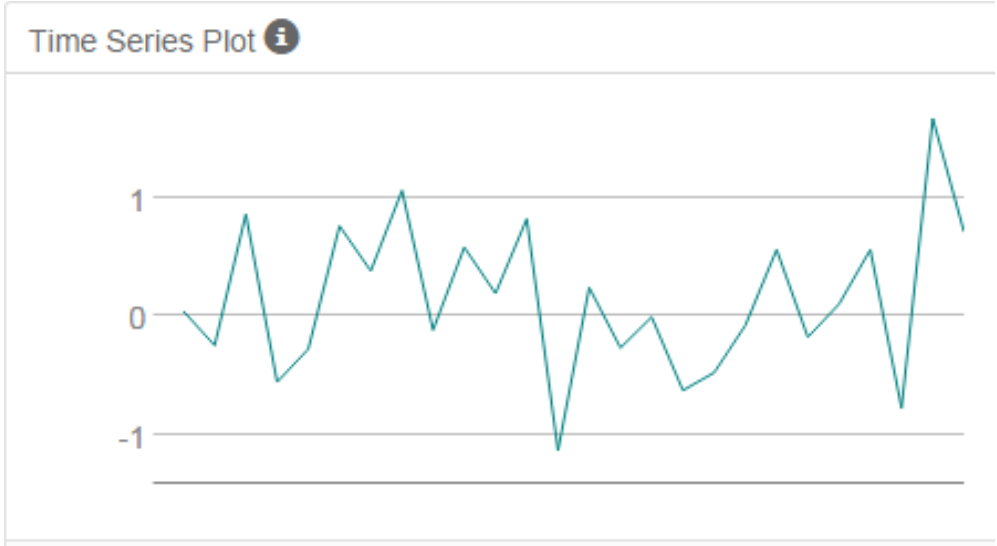
a. Uptrend

b. Downtrend

c. Horizontal Trend

d. Downtrend to Uptrend

e. Uptrend to Downtrend



THANKS FOR YOUR ATTENTION



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